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SAKE: Spatial Question Answering over Knowledge Graph based on Embedding Techniques

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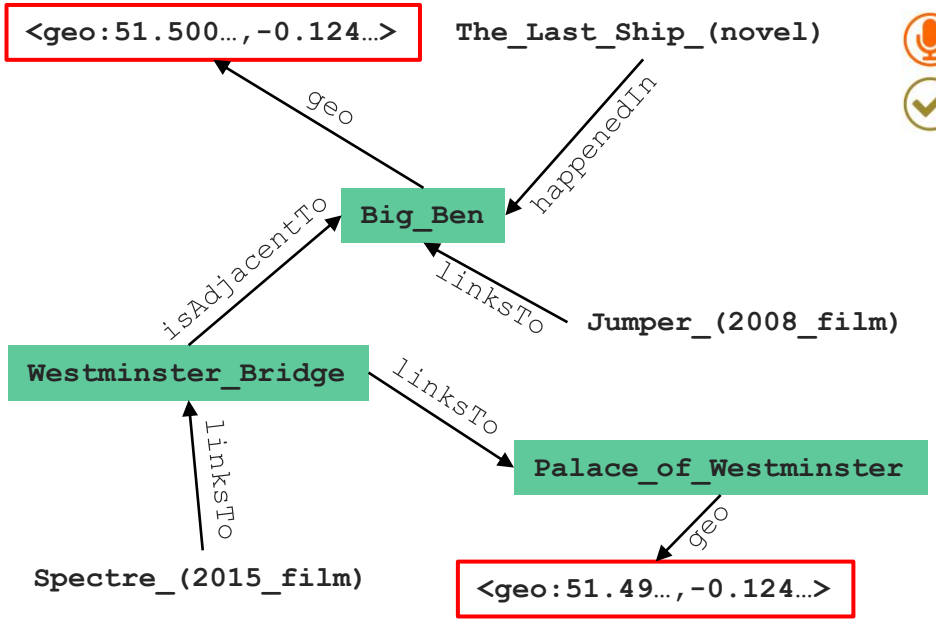
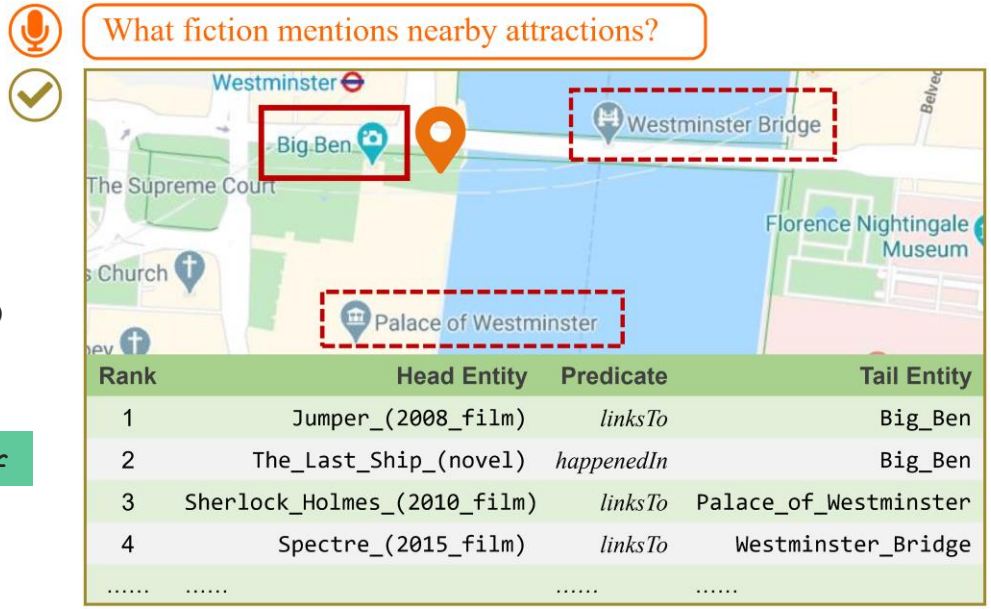
Motivation

- Natural Language **Spatial** Question (NLSQ) answering
  - KGs are increasingly augmented with spatial data/knowledge
  - Spatial awareness to natural language queries in Location-based Services
  - To retrieve *k* spatial facts that are both spatially and semantically relevant





Rank	Head Entity	Predicate	Tail Entity
1	Jumper_(2008_film)	*linksTo*	Big_Ben
2	The_Last_Ship_(novel)	*happenedIn*	Big_Ben
3	Sherlock_Holmes_(2010_film)	*linksTo*	Palace_of_Westminster
4	Spectre_(2015_film)	*linksTo*	Westminster_Bridge
……	……	……	……

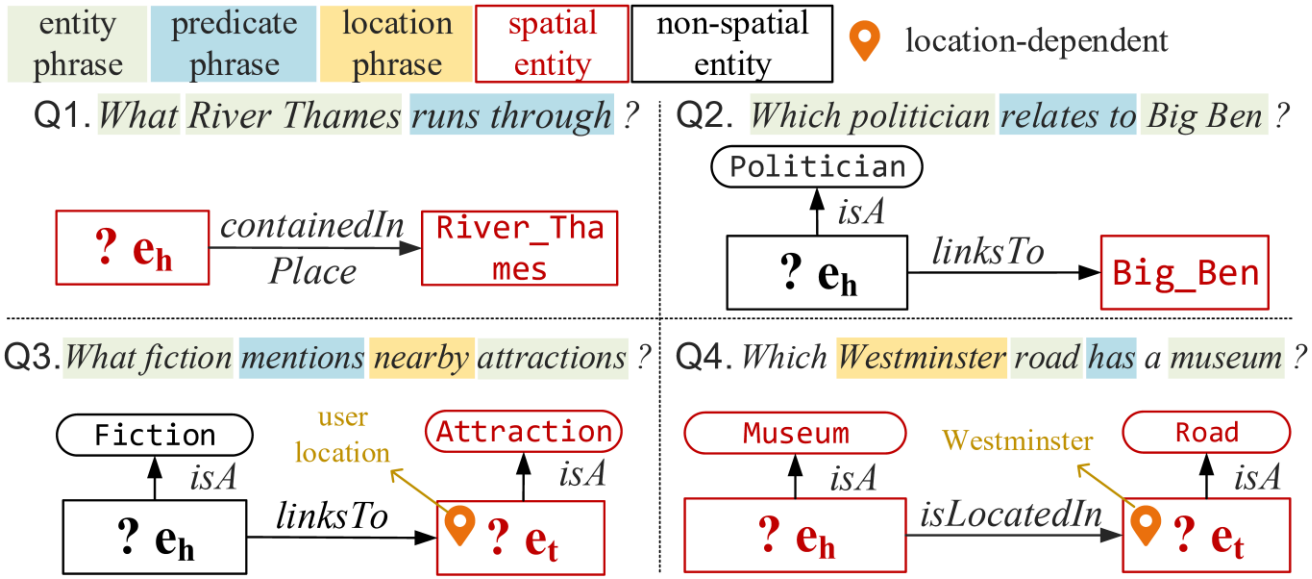
- A traveler to Central London is interested in **nearby** attractions
  - All spatially close entities whose type, attributes, and relations meet the query semantics should be considered and ranked

Four NLSQ Types to be Studied

- A triple $t\ (e_h, \mathrm{r}, e_t)$ is a **spatial fact** if its head or tail entity is a spatial entity
- An NLSQ q^s can be answered by a list of spatial facts. Each of them interprets the answer by matching q^s's mentioned relation and class and being close to q^s's mentioned spatial range if applicable

basic NLSQ asks for some entities having the mentioned relation with the mentioned entity

class-based NLSQ asks for some mentioned class of entities having the mentioned relation with the mentioned entity or class of entities



nearby NLSQ extends a class-based NLSQ such that the asked entities should also be close to the user location

location-name NLSQ extends a class-based NLSQ such that the asked entities should also be close to the location name mentioned by the question

KG Spatial Question Answering

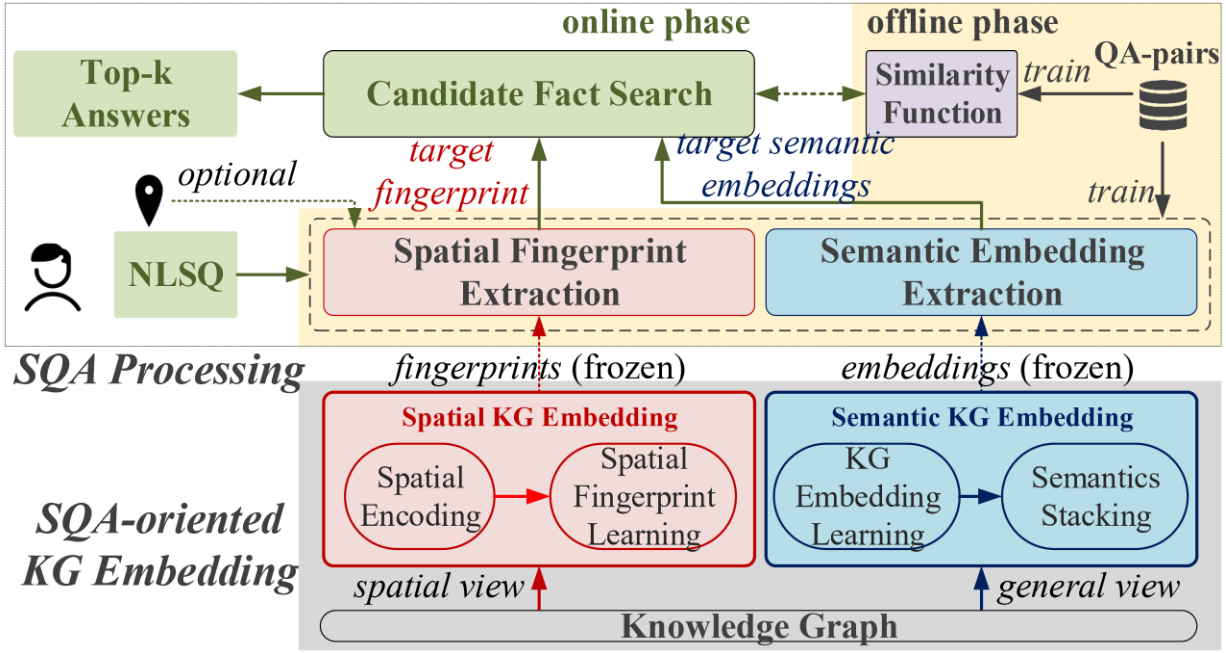
- Given a KG and an NLSQ q^s, KGSQA returns k spatial facts in a set T_q^s , and $\forall\, t \in T_q^s, \Psi(t, q^s) \leq \Psi(t', q^s)$ for any other fact $t' \notin T_q^s$
- $\Psi(t, q^s)$ captures the similarity ranking score of t w.r.t q^s. Three criteria are defined:
  - The spatial entities included in t must match the spatial query intent implied by q^s
  - The entities and predicate of t must be semantically close to their counterparts mentioned in q^s
  - If t is a synthesized fact that that does not pre-exist, its predicted relation of the entities must be reliable and rational
- KGSQA does not target literal values or complex semantics like "Find me *another* museum in Westminster?"

Challenge 1 Data heterogeneity caused by the addition of spatial attributes

Challenge 2 Recognition of spatial constraints

Challenge 3 Integration of spatial closeness and semantic relevance

SAKE Framework



Joint similarity function *is learned to prioritize top-1 fact.*

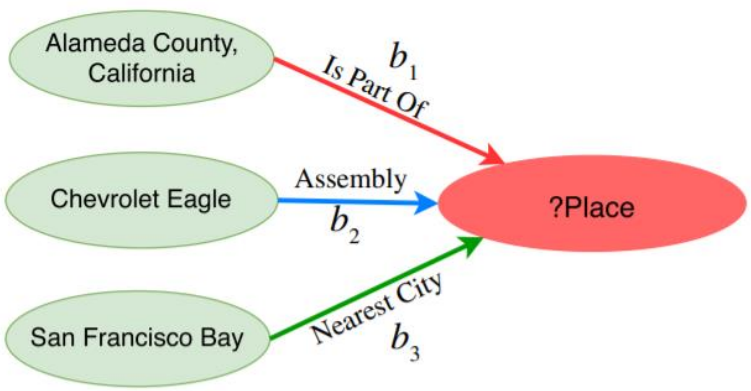
Neural networks *map an NLSQ to frozen spatial fingerprints and semantic embeddings.*

Spatial fingerprints *capture spatial properties of entities as embeddings for matching query intents of NLSQs in embedding spaces.*

- To enable mapping between NLSQ and spatial facts, we utilize KG embeddings
- **SQA-oriented KG Embedding** translates the spatial and semantic information of KG entities/predicates to embeddings
- **SQA Processing** trains models for extracting spatial/semantic embeddings from NLSQs and a similarity function for ranking candidates offline, and retrieves top-*k* answers for an online NLSQ based on the similarity function

SAKE vs. KGQA and GeoQA

- Distance-based KG embedding: $\boldsymbol{e}_t \approx \boldsymbol{e}_h + \boldsymbol{r}$ (TransE) or $\boldsymbol{P}_r\boldsymbol{e}_t \approx \boldsymbol{P}_r\boldsymbol{e}_h + \boldsymbol{r}$ (TransR)
- KGQA: based on similarity function $S(t,q) = ||\mathbf{f}(q), \mathbf{g}(t)||$, $\mathbf{f}$ and $\mathbf{g}$ are two embedding functions for question and KG fact, respectively
- General Procedure of Simple Question Answering over KG:
  1) **Information extraction** from question: LSTM/CNN to extract entity/predicate mentions or embeddings
  2) **Candidate entity/fact selection**: *n*-gram, fuzzy matching, link prediction
  3) **Similarity-based ranking**: string similarity and/or prediction score
- Problems: spatial aspect integration, massive candidates, implicit location name



$$?Place : Is\ Part\ Of\ (Alameda\ County, ?Place) \wedge Assembly\ (Chevrolet\ Eagle, ?Place) \wedge Nearest\ City(San\ Francisco\ Bay, ?Place)$$

Geographical Question Answering

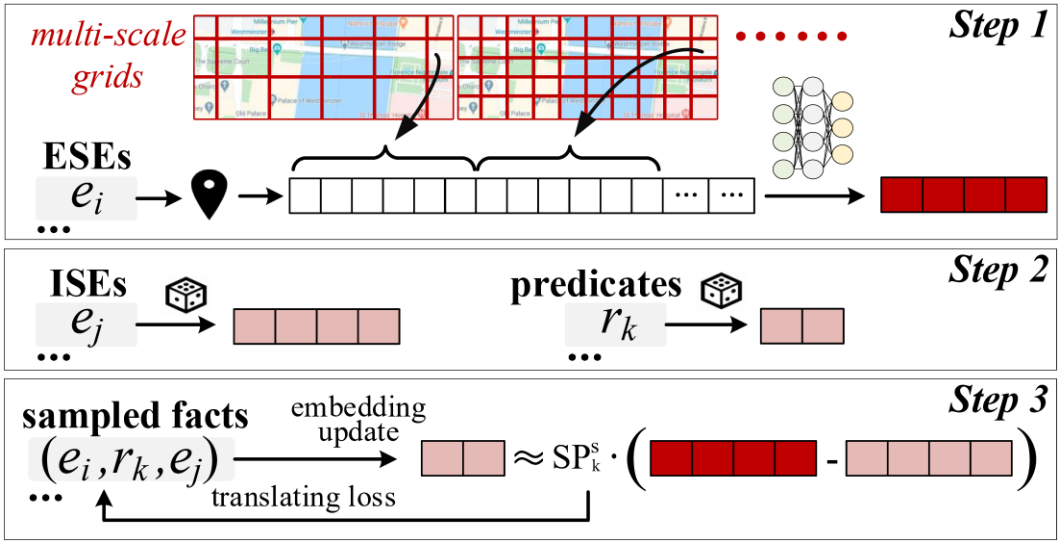
Semantics in the geographical domain, no spatial closeness considered

RDF query-based solution: missing fact issue

A coarser-grained query intents (city-level)

SQA-oriented KG Embedding

Spatial Embedding



Explicit spatial entities (ESEs) link to some geospatial locations or ranges, but **implicit spatial entities (ISEs)** do not: GeoSPARQL fails.

Encode ESEs as spatial fingerprints based on their absolute locations via a multi-scale encoding method.

By utilizing **relations between ESEs and ISEs**, a TransR-based method is used to generate pseudo fingerprints for ISEs.

As a result, any new GPS, geospatial range, and ESEs can be mapped to a spatial fingerprint.

Semantic Embedding

KG embeddings are first generated for entities and predicates based on TransR.

A semantics stacking is performed by 1) embedding labels and words of entities/predicates, 2) append textual embeddings to the generated KG embeddings.

Entity `Big_Ben` *-> KG labels ['Big Ben', 'Elizabeth Tower', ...] -> mean of GloVe word embeddings*

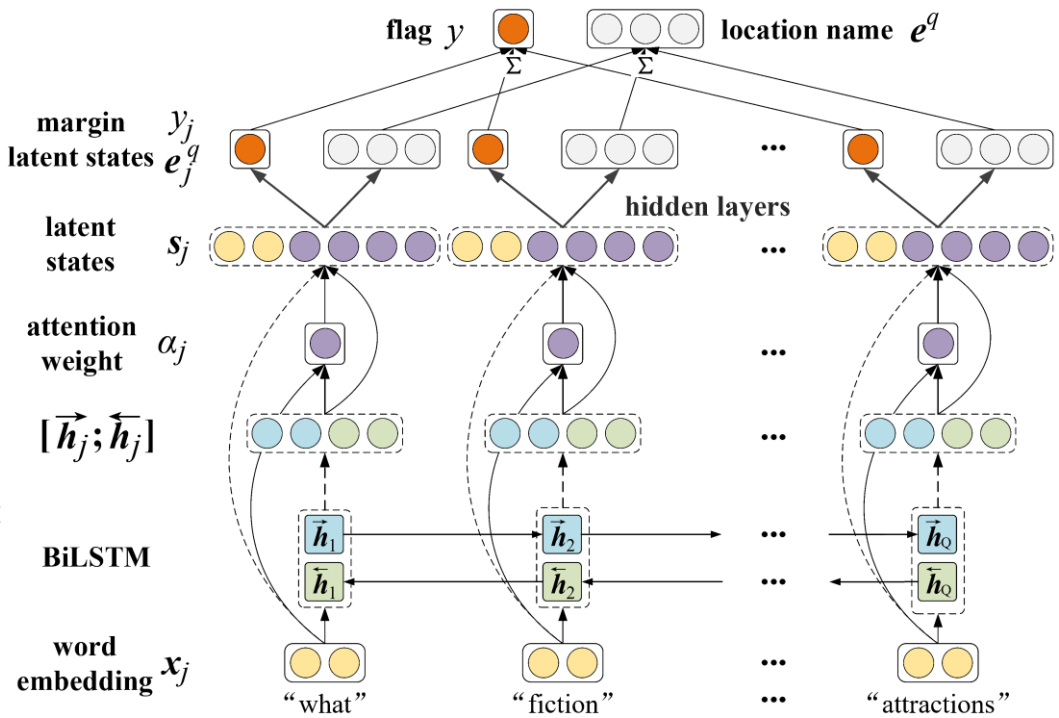
NLSQ Extraction Models

Fingerprint Extraction

Extracted spatial fingerprint serves as the **spatial query intents of the NLSQ**.

Cases are differentiated for nearby NLSQs and others: $y \in \{0, 1\}$. For nearby NLSQ, user location is encoded as fingerprint.

For others, a spatial entity whose semantic embedding is closet to the predicted semantic embedding provides its fingerprint for candidate search.



Semantic Embedding Extraction for Mentioned Entity/Predicate

Using a similar idea to fingerprint extraction, an NLSQ are mapped to frozen semantic embeddings learned for the KG entity and predicate mentioned by the NLSQ.

Our intuition of learning such mapping is that the target entity/predicate's information (e.g., name/alias, contextual entities, and entity type) has been captured in both NLSQ and the output embedding .

Similarity Function & Fact Search

Similarity Function Modeling and Learning

Semantic Relevance: semantic embedding similarity between the target head entity (predicate) and its candidate + predicted relation reliability in terms of embedding translating loss.

$$\Psi(t, q^s) = \Psi^{\text{sm}}(t, q^s) + \Psi^{\text{sp}}(t, q^s) \tag{5}$$

$$\Psi^{\text{sm}}(t, q^s) = \lambda_{\text{h}}||\mathbf{e}_h - \hat{\mathbf{e}}_h||_2 + \lambda_{\text{r}}||\mathbf{r}_o - \hat{\mathbf{r}}_o||_2 + \lambda_{\text{t}}||\mathbf{r}_o^\star - \mathbf{P}_o^r(\mathbf{e}_h^\star - \mathbf{e}_t^\star)||_2$$

Spatial Closeness: translating loss in spatial fingerprint space + fact's consistency with the spatial query intent.

$$\Psi^{\text{sp}}(t, q^s) = \lambda_{\text{sr}} \min(\mathrm{I}(e_t), \mathrm{I}(e_h))||\mathbf{sr}_o - \mathrm{SP}_o^r(\mathbf{se}_h - \mathbf{se}_t)||_2 + \lambda_{\text{se}} \min(\mathrm{I}'(e_h)||\mathbf{se}_h - \hat{\mathbf{se}}^q||_2, \mathrm{I}'(e_t)||\mathbf{se}_t - \hat{\mathbf{se}}^q||_2)$$

Head Spatial Fact (SF), Tail SF, and Full SF are integrated into the design by **indicator functions**.

Margin-based ranking loss, negative sampling, grid search for parameter combination.

$$\min_{\lambda_{\text{t}}, \lambda_{\text{r}}, \lambda_{\text{h}}, \lambda_{\text{se}}, \lambda_{\text{sr}}} \sum_{q^s \in \mathcal{Q}} [\Delta' - \Psi(t_g, q^s) + \frac{1}{K} \sum_{t' \in \mathcal{C}^K} \Psi(t', q^s)]_+$$

Candidate Fact Search Algorithm

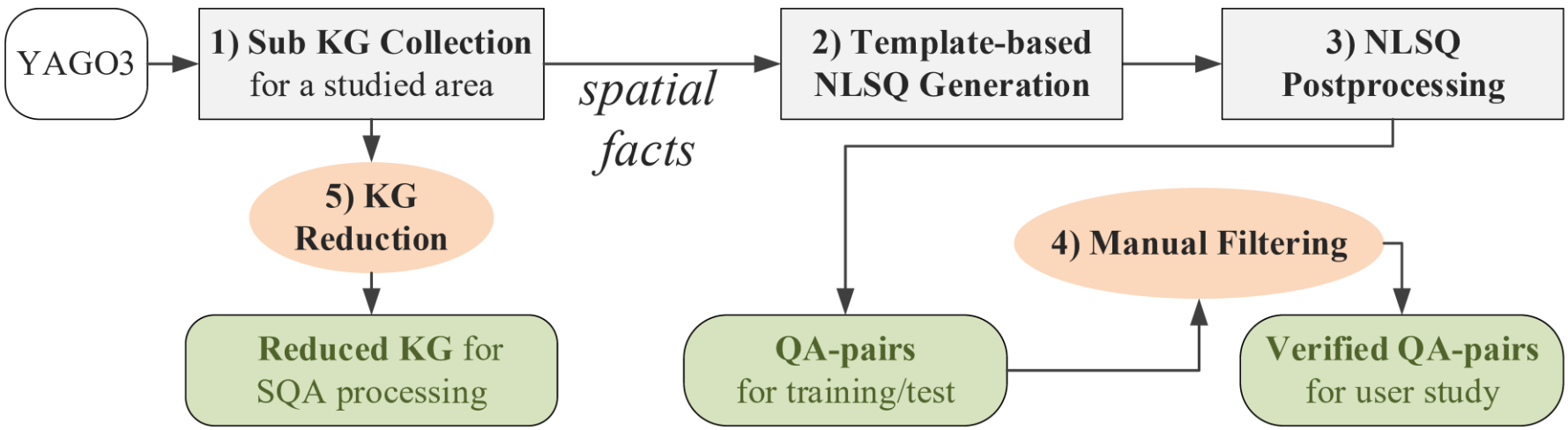
3: $CSE \leftarrow$ find kNN spatial entities of $\hat{\mathbf{se}}^q$ on KD^{sp}
4: put $\langle e_x, ||\mathbf{se}_x, \hat{\mathbf{se}}^q||_2 \rangle$ in $\mathcal{M}_{\text{se}}$ for each $e_x \in CSE$
5: $CR \leftarrow \{r_j \mid ||\mathbf{r}_j, \hat{\mathbf{r}}_o||_2 \leq \eta^r\}$
6: put $\langle r_j, ||\mathbf{r}_j, \hat{\mathbf{r}}_o||_2 \rangle$ in $\mathcal{M}_{\text{r}}$ for each $r_j \in CR$

34: **if** $\mathcal{F}.has(flag)$ and $|\mathcal{C}'| > k'$ **then**
35: compute each fact's similarity score by Equation 5
36: $\mathcal{C}'.\Psi_{k'} \leftarrow$ the sum of top-k' lowest scores
37: **if** $\mathcal{C}'.\Psi_{k'} > \Psi_{k'}$ of other category **then** $\mathcal{F} \leftarrow \mathcal{F} \setminus flag$

KD-tree, beam search, greedy

top sampling, early termination, 3 categories

Datasets



- Studied areas: Beijing (BEJ), Copenhagen (CPH), London (LON)
- Generate NLSQs based on templates and grammatical transformation

Datasets	LON	CPH	BEJ
# of spatial facts	516.9k	35.7k	35.2k
# of basic NLSQs	186.3k	21.5k	15.5k
# of class-based NLSQs	1.03M	71.4k	70.3k
# of nearby NLSQs	969.9k	45.8k	32.7k
# of location-name NLSQs	688.6k	29.6k	20.2k
# of QA-pairs	2.87M	168.3k	138.7k

- Manual Filtering to obtain verified QA-pairs (~1k per method per each dataset)
  - **Rephrasing (^{R})**: multiple volunteers vote and NLSQ is rephrased for a supported QA-pair
  - **Mapping (^{M})**: map a QA-pair to a FB human question set according to fact structure
- Remove KG facts to test model's ability to synthesize missing facts
- https://github.com/sake-project/sake-preprocessing

SQA Performance

TABLE III: Performance Comparison of SQA Methods (best results are in bold)

Dataset	LON	CPH	BEJ	LON^R		CPH^R		BEJ^R		LON^M		CPH^M		BEJ^M	
Metric (unit: %)	R@1	R@1	R@1	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5
CQAA	30.9	36.7	51.9	41.4	50.2	41.6	50.6	47.2	57.3	31.6	39.8	37.1	51.8	49.7	65.1
BuboQA	34.2	43.6	54.3	41.8	51.1	56.8	65.5	48.3	59.1	35.2	42.1	43.1	54.7	52.3	66.9
UBSQA	34.9	43.8	54.8	42.4	51.3	56.9	65.5	48.5	59.6	36.4	44.4	43.7	54.9	52.5	68.2
KGEQA	38.9	44.7	55.3	44.8	53.2	57.4	66.9	48.8	59.6	39.5	47.1	44.5	55.9	53.6	68.8
BERTQA	38.7	44.8	55.7	44.9	53.0	57.2	66.6	48.9	60.2	39.8	46.6	44.1	55.4	53.3	68.8
SAKE\S	37.8	42.2	53.2	43.2	51.9	56.2	65.3	48.5	57.8	38.3	45.9	43.5	54.2	52.4	67.7
CQAA+S	59.7	64.4	69.7	69.8	77.3	60.2	69.4	59.2	68.9	60.3	70.4	64.7	71.5	68.3	78.2
BuboQA+S	66.5	69.6	71.8	70.1	78.1	75.1	81.8	68.6	78.7	67.4	77.2	69.5	79.6	69.5	79.4
UBSQA+S	68.2	70.2	72.6	71.8	79.7	75.3	82.3	68.8	78.9	68.5	78.4	70.1	80.5	70.8	80.9
KGEQA+S	71.3	70.6	75.1	73.8	81.2	75.8	83.4	70.2	79.4	71.6	79.2	70.7	80.6	74.2	83.2
BERTQA+S	70.3	70.2	73.0	72.1	80.5	75.3	82.6	68.9	79.4	70.5	78.7	70.1	80.3	71.3	81.6
SAKE (ours)	**74.2**	**78.5**	**77.6**	**76.6**	**83.3**	**77.8**	**84.4**	**72.9**	**81.0**	**75.3**	**79.8**	**78.2**	**83.3**	**76.3**	**84.1**
SAKE\GloVe	60.3	67.7	59.8	67.6	74.3	74.4	79.9	56.5	64.0	59.1	62.6	61.3	66.4	57.4	65.1

- Five KGQA approaches, original or enhanced with spatial fingerprints (+S)
  - Use spatial fingerprints to expand candidates, and linearly combine the original similarity score and our fingerprint-based score (tuned optimal)
- R@*k*, i.e., whether the closest fact appears in the top-*k* results of an NLSQ
- Similarity enabled by spatial fingerprints is crucial (original vs. +S version)
- Semantics stacking makes a good supplement (SAKE vs. SAKE\Glove)
- SAKE excels at synthesizing missing facts and utilizing joint similarity (SAKE vs. +S alternatives)
- SAKE stay relatively stable on all human question datasets ($*^R$ and $*^M$)

Selected Evaluation Results

TABLE VI: Recall of Tiananmen's GPS Point on BEJ

#	Entity Name	Distance (m)	True Rank
1	Tiananmen	0.0	1
2	Forbidden_City_Concert_Hall	558.9	7
3	Zhongshan_Park_(Beijing)	259.6	3
4	Xinhua_News_Agency	2402.2	90
5	Gate_of_China_(Beijing)	693.6	8
6	Collections_of_the_Palace_Museum	919.5	11
7	Meridian_Gate	574.5	7
8	Tiananmen_Square	447.5	6
9	Xuanwu_District,_Beijing	1630.3	45
10	Beijing_Shejitan	415.7	5

Spatial Fingerprints

use Tiananmen's GPS to search for its NN entities in the fingerprint space.

10NNs are all within 2.5 km. 7 out of 10 returned are in ground truth (sufficient for subsequent search).

ISEs without GPS can be recalled as well.

NLSQ Types

SAKE has clear advantages on nearby and location-name NLSQs.

SAKE works better than or equally to two SOTA methods in basic and class-based NLSQs, as it adopts embeddings full-stack to capture NLSQ semantics.

Deep integration of spatial and semantic similarities.

TABLE IV: Rephrased NLSQ Examples of a Spatial Fact in CPH^R

A fact: (Sven_Risom, *created*, St._Anne's_Church,_Copenhagen)

Basic: *Who builds the St. Anne's Church in Copenhagen?*
Class-based: *Which European people is the Sankt Annæ Kirke's creator?*
Nearby: *Which neighboring church is created by a Dane?* #with user location (55.6559, 12.6082)#
Location-name: *What building in København S district is built by Sven Risom?*

TABLE V: Result Effectiveness vs. NLSQ Types in CPH^R

NLSQ types	# of QA-pairs	R@1 (unit: %) SAKE	BERTQA	KGEQA+S	BERTQA+S
basic	22	**81.8**	77.3	**81.8**	**81.8**
class-based	369	**82.7**	82.1	82.1	82.4
nearby	361	**72.9**	30.7	64.5	59.3
location-name	277	**77.6**	57.0	74.0	74.4

Summary

- Four typical NLSQ types in one framework
- SAKE learns the embeddings of KG entities/predicates in two processes for spatial and semantic properties, respectively
- SAKE maps natural language questions to the learned embeddings of their mentioned entities and predicates
- SAKE trains joint-similarity function in the embedding space and couples it with a candidate fact search algorithm

SAKE Project

- Future directions
  - Multi-task learning to integrate KG embedding and SQA
  - Spatial KG embeddings for location-based recommendations
  - More complex NLSQs, more advanced NLP and embedding techniques